

DBMS Unit 3 — Normalization & Schema Refinement Complete Notes

Database Management Systems (DBMS) · Unit 3 · Normalization & Schema Refinement · 2025

Overview

These notes cover **Normalization & Schema Refinement**. Design, query and manage relational databases — from ER modelling and SQL to normalization, transactions and recovery. Each section states the core idea, a worked intuition and the points examiners look for.

1. Functional Dependencies

A functional dependency $X \rightarrow Y$ states that tuples agreeing on X must agree on Y ; Armstrong's axioms derive all implied dependencies.

Why it matters: Functional Dependencies appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of functional dependencies in one line.
- A small example showing functional dependencies in practice.
- Common mistakes students make and how to avoid them.

2. Closure of Attributes

The attribute closure X^+ is used to test candidate keys and to check whether a dependency is implied.

Why it matters: Closure of Attributes appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of closure of attributes in one line.
- A small example showing closure of attributes in practice.
- Common mistakes students make and how to avoid them.

3. First, Second and Third Normal Form

1NF requires atomic values, 2NF removes partial dependencies on a key and 3NF removes transitive dependencies.

Why it matters: First, Second and Third Normal Form appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of first, second and third normal form in one line.
- A small example showing first, second and third normal form in practice.
- Common mistakes students make and how to avoid them.

4. Boyce-Codd Normal Form

BCNF requires every determinant to be a super key; it is stricter than 3NF and may not preserve dependencies.

Why it matters: Boyce-Codd Normal Form appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of boyce-codd normal form in one line.
- A small example showing boyce-codd normal form in practice.
- Common mistakes students make and how to avoid them.

5. Lossless Join Decomposition

A decomposition is lossless if the natural join of the parts reproduces the original relation exactly.

Why it matters: Lossless Join Decomposition appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of lossless join decomposition in one line.
- A small example showing lossless join decomposition in practice.
- Common mistakes students make and how to avoid them.

6. Multivalued Dependencies and 4NF

Multivalued dependencies capture independent multi-valued facts; 4NF eliminates non-trivial MVDs.

Why it matters: Multivalued Dependencies and 4NF appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of multivalued dependencies and 4nf in one line.
- A small example showing multivalued dependencies and 4nf in practice.
- Common mistakes students make and how to avoid them.

Quick revision

Functional Dependencies · Closure of Attributes · First, Second and Third Normal Form · Boyce-Codd Normal Form · Lossless Join Decomposition · Multivalued Dependencies and 4NF